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A TOTAL ACADEMIC AND RESEARCH SOLUTIONS PROVIDER

A WORKSHOP ON

K-CHART™:

AN EFFECTIVE TOOL FOR RESEARCH PLANNING AND MANAGEMENT

K&Q ANALYTICS SDN. BHD.

Name : ______________________________

Institution :_____________________________

WORKSHOP OUTLINE

INTRODUCTION

ISSUES IN RESEARCH, DEVELOPMENT, COMMERCIALIZATION AND BUSINESS (RDCB) VALUE CHAINS

WHY K-CHART

WHAT IS K-CHART

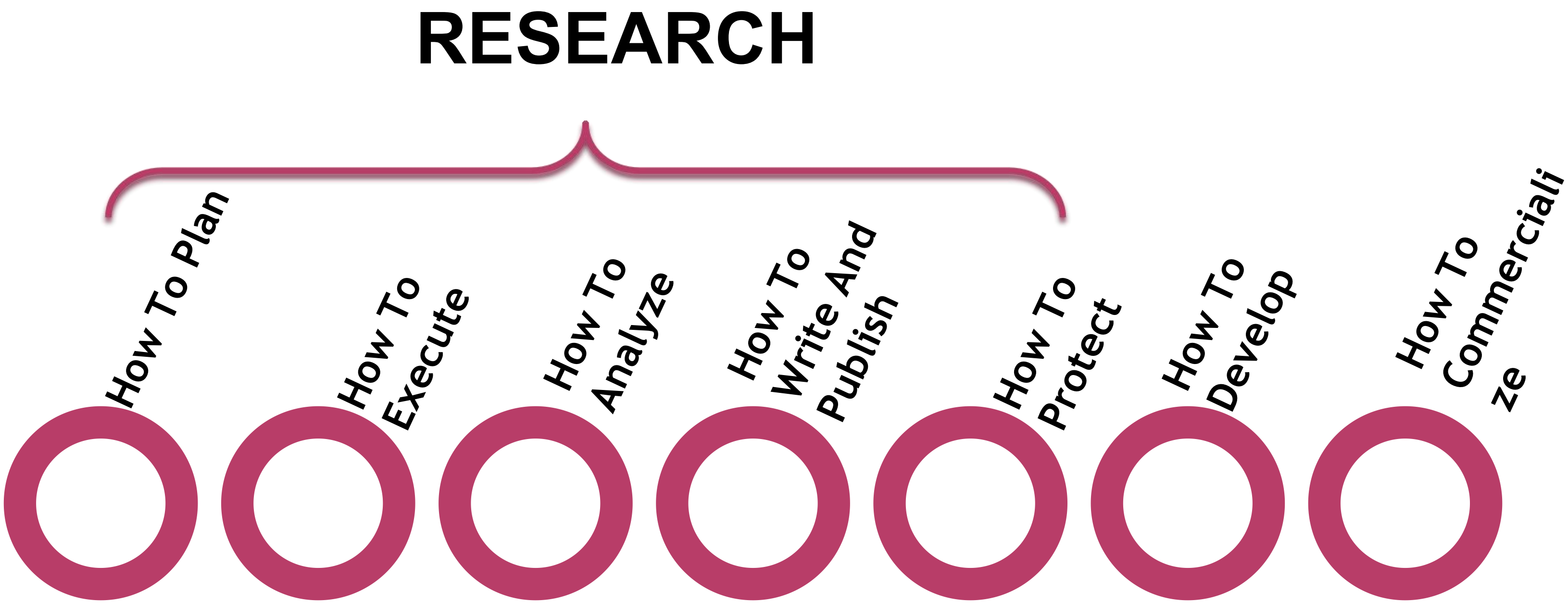
CONSTRUCTION OF K-CHART

SELF PRACTICE: CONSTRUCT YOUR OWN K-CHART

SLOT 1:
BACKGROUND: SOME FUNDAMENTAL ISSUES IN RESEARCH

1. What is research
2. Distinguishing Research from Development
3. Objective of Research (Spiral of Knowledge)
4. Types of research
5. Quality of research

TRAINING (KNOWLEDGE & SKILL) NEEDS



EXAMPLES OF RESEARCH OBJECTIVES

OBJECTIVE	EXAMPLE	ANALYSIS
Collecting Data	Population/prevalence/frequency (of people, animals, plants, astronomical bodies, companies, diseases etc.)	DV is Population (number) IV is the type of subject Descriptive Analysis
	Trend/Pattern (geographical/spatial, temporal)	DV: Population (number) IV: Subject, Time, Place Descriptive, Relationship Anal.
Understanding/ Identification of Strengths & Weaknesses	Scoring/Indexing/ranking/rating (Innovation Capabilities, Economics-GDP, CPI, GPI, Productivity; Quality Index-Air, Water, etc.)	DV: Aggregated Parameter IV: Indicators, Parameters Descriptive, Relationship Anal., Factor Anal.
Solution	to increase efficiency •Increase Productivity/Cost •Cost- materials, resources, time etc •Relates to quantity	DV: Performance Parameters IV: Design Parameters Causal Relationship Analysis, Comparative Analysis
	to increase effectiveness •Performance •Relates to quality	As above
Implementation	•To get real feedbacks and responses •To improve for the next solutions/products/policies	DV: Performance Parameters IV- Not important

TYPES OF RESEARCH

- **TECHNICAL**
  - Pure Sciences
  - Engineering
  - Technologies
- **NON-TECHNICAL**
  - Humanities/ Soc Sciences
    - System
    - Technique/Process/Phenomena
      - Qualitative
      - Quantitative
        - Exploratory
        - Fundamental
        - Applied
          - Incremental
            - Non-participatory/ Conventional
              - Primary
              - Secondary
            - Participatory/Action
          - Radical
        - DEVELOPMENT
      - Hybrid
  - Economy/Management/ Business/Finance

RESEARCH QUALITY

Defining quality of research:

Using LOWEST COST;

- SHORTEST TIME
- SIMPLEST METHOD
- LEAST RESOURCES

R_{INPUT}

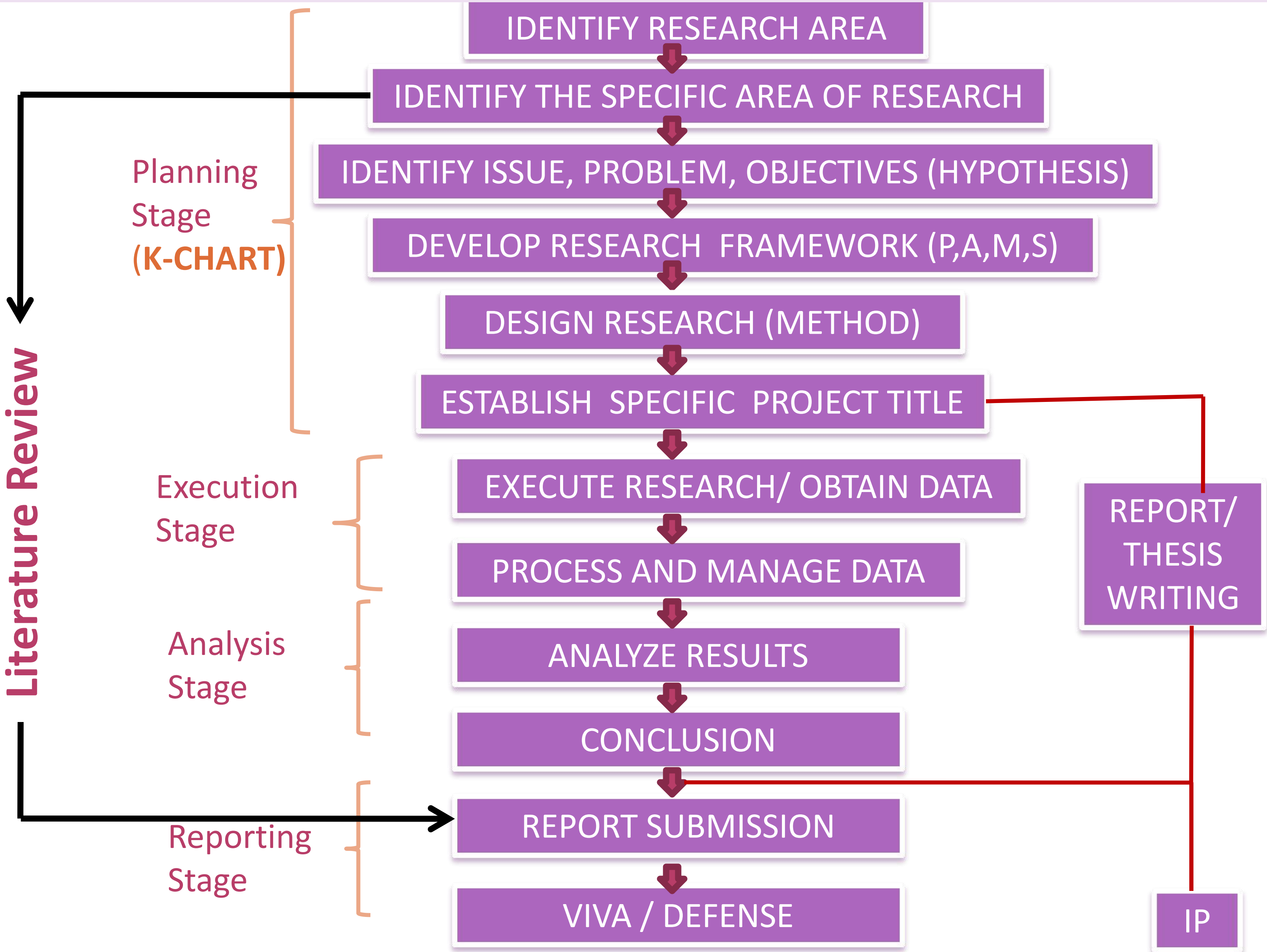
To obtain BIGGEST IMPACT

- WIDEST APPLICATION
- MOST CRITICALLY NEEDED
- BEST PERFORMANCE
- MOST COMPLETE
- MOST NOVEL/INNOVATIVE

R_{OUTPUT}

RESEARCH GAIN, $R_{GAIN} = R_{OUTPUT} / R_{INPUT}$

TYPICAL RESEARCH FLOW



SLOT 2:

WHY K-CHART™?

GRADUATE STUDY SCENARIO
RESEARCH PROJECT SCENARIO

1. SCENARIO 1: GRADUATE STUDIES
2. SCENARIO 2: EMBARKING ON A RESEARCH PROJECT
3. QUESTIONS IN RESEARCH PLANNING
4. **STRUCTURE OF A K-CHART™**
5. BENEFITS OF K CHART IN RESEARCH PLANNING & MANAGEMENT
6. EXAMPLES FROM THE K CHART FOR VEHICLE

SCENARIO 1: GRADUATE STUDIES

- You just started your graduate study. You are looking for a title. Your supervisor is always busy.
- You pick up and photocopy every single article you come across, only to sleep on it than to read it through.
- Even when you do finish an article, it's like reading Gibberish. You don't know what to get from the paper.
- You keep asking yourself; What exactly am I working on? How much should I read? What should be my methodology? How many results should I have? How do I get a novelty?

SCENARIO 2: EMBARKING ON A RESEARCH PROJECT

You are about to embark on a research project. You face a hard time identifying the **scope** of research and **specific problems** to solve.

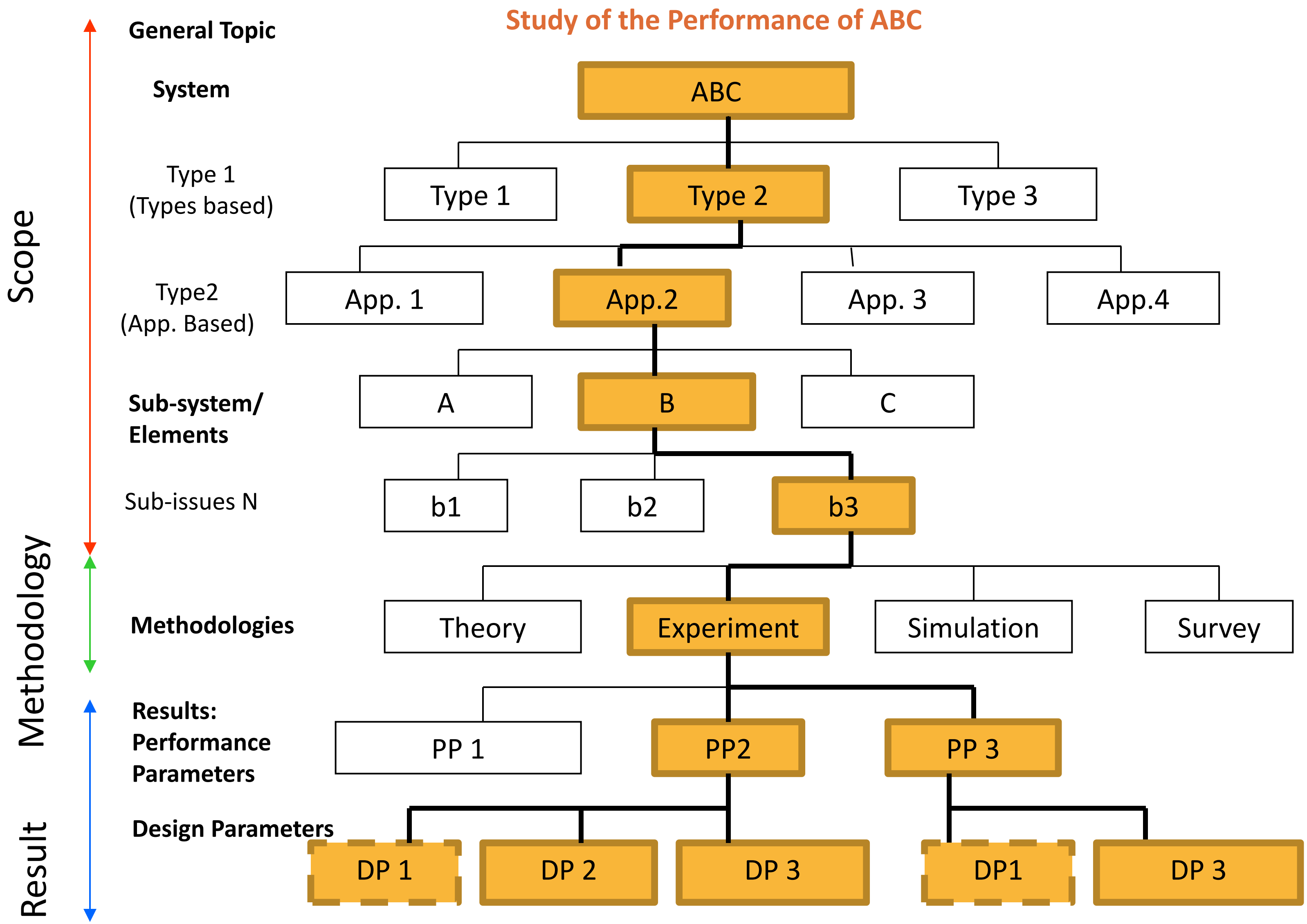
You are planning to have co-researchers/ students, but **organizing and coordinating** them are not a piece of cake. You want them to work together, not to scream at each other.

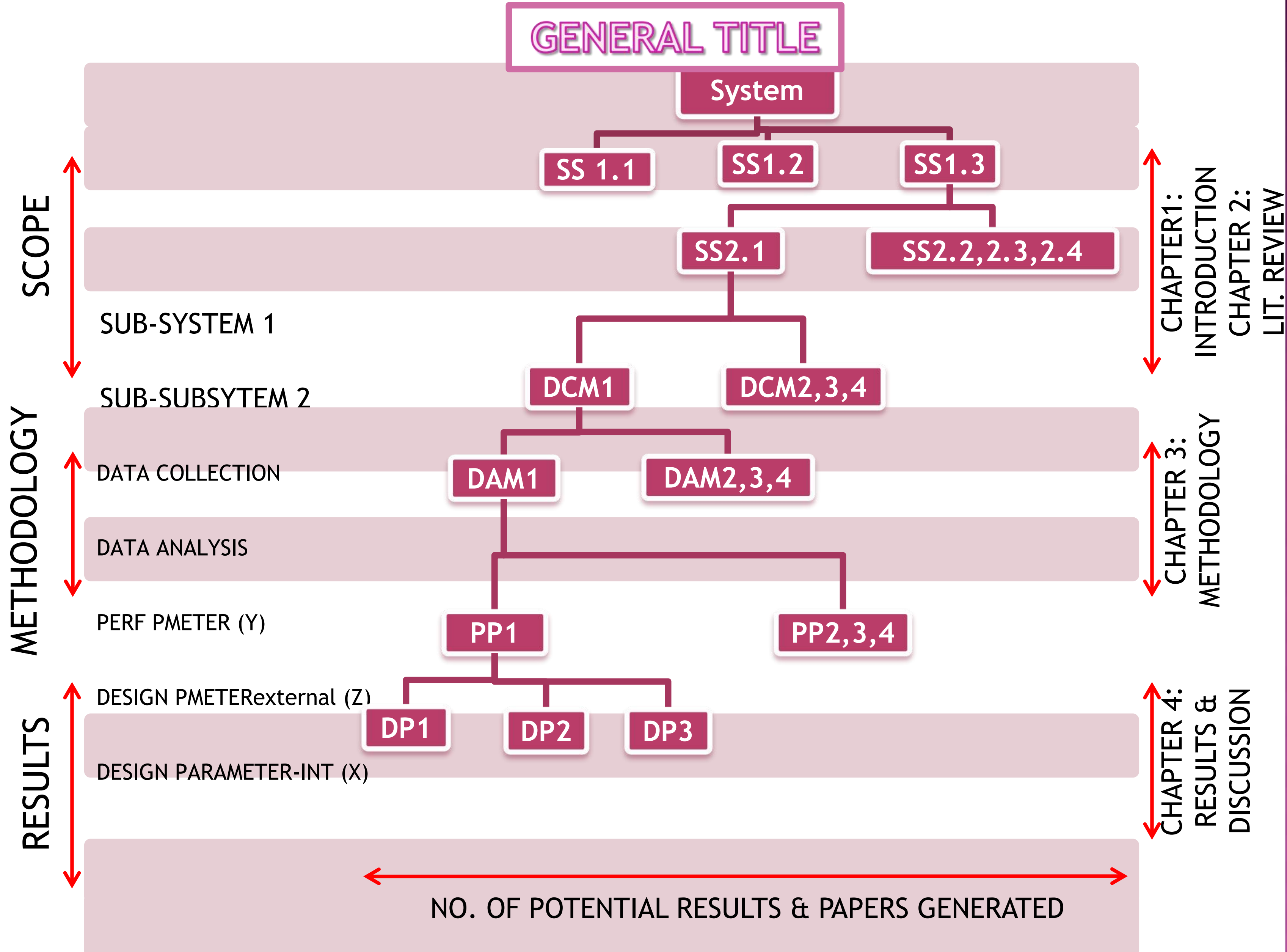
You ask yourself; who will do what, and how much? How many sub-projects can be created? How to maximize the outputs? How to produce more papers?

QUESTIONS IN RESEARCH PLANNING

1. What is the specific title of study
2. How to identify the problems to study?
3. How wide/deep is the study?
4. How much is the literature review?
5. What do I look for in reading reference papers ?
6. How do I choose between two (or more) design options?
7. How do I organize my study?
8. Should I focus on theory or simulation or what?
9. How many results should I get?
10. How do I know whether my results are correct and sufficient?
11. What if I can not get the target results?
12. How to arrange the results?
13. How to cluster for sub-projects?

STRUCTURE OF A K-CHART™




GENERAL TITLE
System
SS 1.1
SS1.2
SS1.3
SS2.1
SS2.2,2.3,2.4
DCM1
DCM2,3,4
DAM1
DAM2,3,4
PP1
PP2,3,4
DP1
DP2
DP3
SCOPE
METHODOLOGY
RESULTS
SUB-SYSTEM 1
SUB-SUBSYTEM 2
DATA COLLECTION
DATA ANALYSIS
PERF PMETER (Y)
DESIGN PMETERexternal (Z)
DESIGN PARAMETER-INT (X)
NO. OF POTENTIAL RESULTS & PAPERS GENERATED
CHAPTER1: INTRODUCTION
CHAPTER 2: LIT. REVIEW
CHAPTER 3: METHODOLOGY
CHAPTER 4: RESULTS & DISCUSSION

BENEFITS OF K CHART IN RESEARCH PLANNING & MANAGEMENT

Identify the scope of study

Guide for Literature Review (directed reading by knowing what you do not know)

Guide in choosing the main issues under study. Distinguish the focused issues from complementary ones

Gauge the level of quality of the research project, thus the potential for novelties and publications

Identify the level of assumptions

Establish the problems and objectives of research

Identify the methodology

Identify the results

MORE BENEFITS

Guide to attain quality research through the selection criteria (zoom & expand)

Distinguish Research projects (PP and DP, more analyses) from Development projects (only PP, no DP, specifications rather than analyses). The less the Results layer (the PP and DP), the smaller the R, the larger the D

Distinguish Pure Fundamental research from Pure Applied research. The more fundamental the research, the more general (the wider) the Scope layer

with a K-CHART™, let's try to answer these questions again

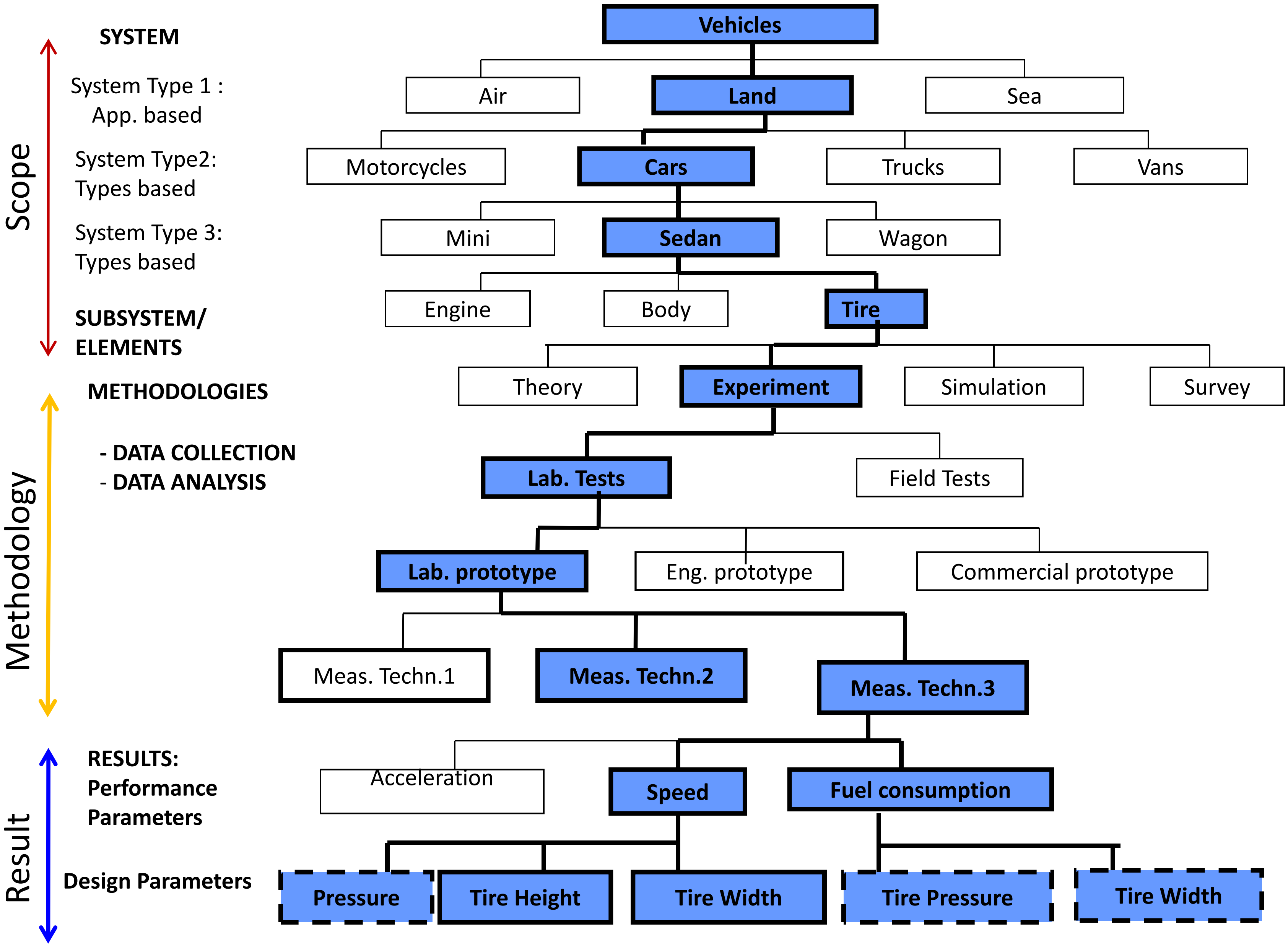
1. What is the specific title of study
2. How to identify the problems to study?
3. How wide/deep is the study?
4. How much is the literature review?
5. What do I look for in reading reference papers ?
6. How do I choose between two (or more) design options?
7. How do I organize my study?
8. Should I focus on theory or simulation or what?
9. How many results should I get?
10. How do I know whether my results are correct and sufficient?
11. What if I can not get the target results?
12. How to arrange the results?
13. How to cluster for sub-projects?

and even more questions…

15. How to plan for publication ?
16. How to construct articles' titles ?
17. How to organize the report structure ?
18. How to plan for a graduate thesis writing?
19. How to integrate with new ideas ?
20. How to re-align the area of research when the original plan does not work ?

GENERAL TITLE

EXAMPLE: Factors Influencing the Performance of Motor Vehicles



EXAMPLES FROM THE K CHART FOR VEHICLE

- Project Title:
- Chapter Title:
- Paper Titles:
- Literature Reviews:
- Assumptions:
- Number of Results:

SLOT 3:

WHAT IS K-CHART™?

1. MAIN ISSUES IN RESEARCH: S.P.A,M
2. FRAMEWORK OF A RESEARCH PROJECT: IDENTIFY SPAM – ORGANIZE YOUR THOUGHTS
3. WHAT IS A K-CHART™?
4. THE K-CHART™ LAYERS

S.P.A.M

Main Issues in Research:

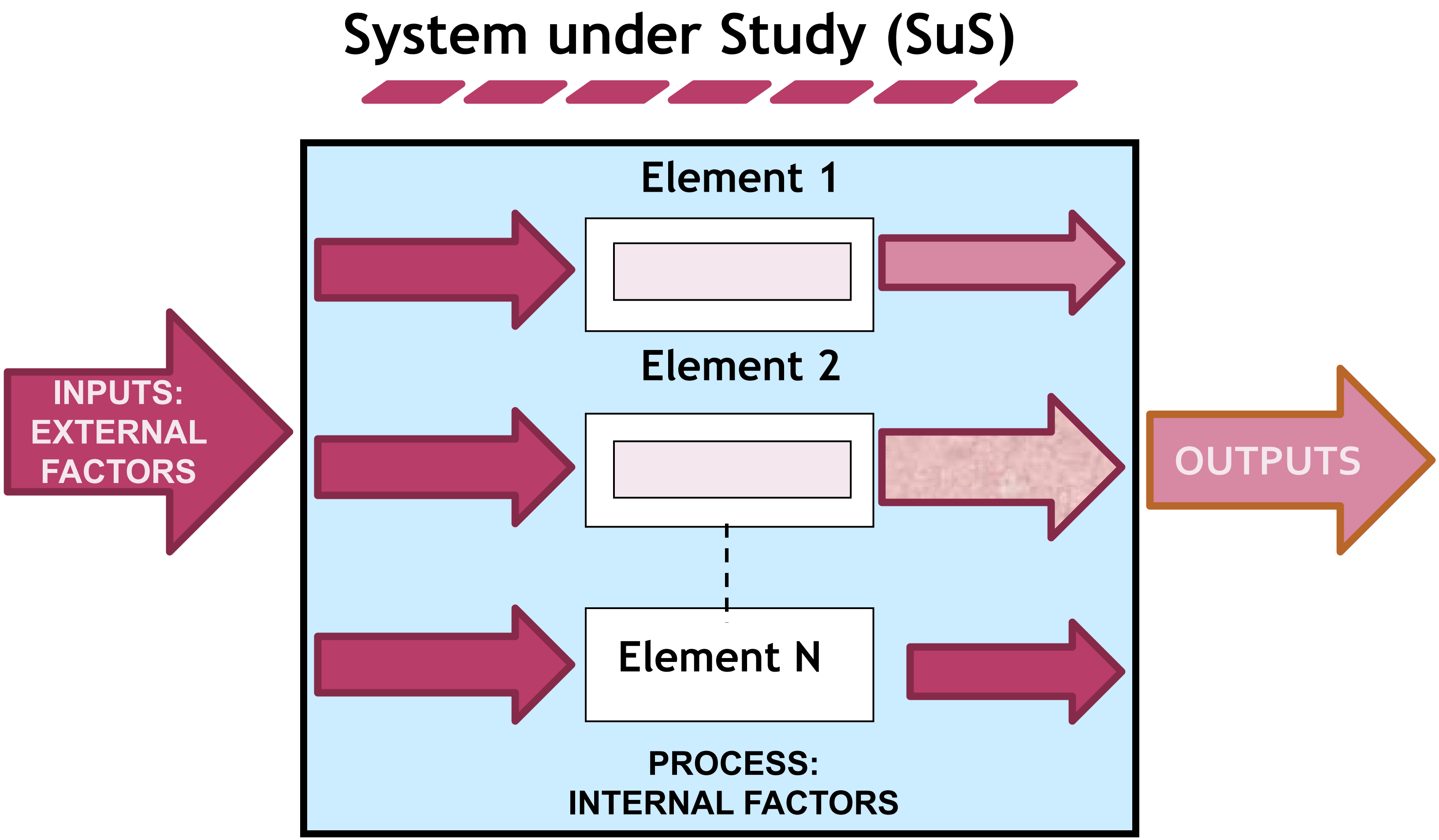
S
- **System** under study

P
- **Problems** to be solved

A
- **Achievement** in solving the problems

M
- **Method** used to achieve the solutions

FRAMEWORK OF A RESEARCH PROJECT: IDENTIFY SPAM



WHAT IS A K-CHART™?

A tool to help establish SPAM

By systematically organizing;

- Scope of Study
- Problems under research
- Methodology
- Results
- Timelines

In the form of a Tree Diagram

Through the process of "Zoom and Expand"

With specific rules

THE K-CHART™ LAYERS

GENERAL TITLE

↓

PROBLEMS

SCOPE

↓

METHODOLOGY

↓

RESULTS (ACHIEVEMENT)

↓

TIMELINES

SLOT 4:

CONSTRUCTION OF K-CHART™?

STEPS IN CONSTRUCTING A K-CHART

STEPS	ACTIVITY
1	Constructing the general title
2	Constructing the Scope of System under study layers
3	Constructing the Methodology layers
4	Constructing the Result layers
5	Constructing the Timelines

Note: The layers can be initially constructed independently

STEP 1: CONSTRUCTING THE TITLE

- A title should indicate:
  - The **S**ystem you are working on
  - The **P**roblems you are solving
  - Your **A**chievements
  - Your **M**ethodology

- General title: SPAM in general (no achievements)
- Program/Project Title: Specific SPAM
- Paper Title: Very Specific SPAM

CONSTRUCTING THE GENERAL TITLE

One normally begins a research project with a general idea of what he/she wants to do

The general idea normally appears in the forms of the issues or the problems he/she wants to study in a certain area of research

So a research idea normally begins with issues/problems (P), while the System (S), the Method (M) and the Achievements (A) are not well formed yet.

However, quite often that Achievements (A) are directly identifiable through Problems (P)

EXAMPLES OF TITLES

- GENERAL TITLE:
  - Factors affecting the performance of motor vehicles
  - Customer Satisfaction in Tourism Industry
  - Development of an Index for Human Innovativeness
  - Dynamics of Firefly Population at Pjaya Wetlands
  - Pattern of spread of Contagious Diseases in Malaysia
  - Patients’ Satisfaction towards Services in General Hospitals
- PROJECT TITLE: Patients’ Satisfaction towards Out-patient Services in General Hospitals
- PAPER TITLE: Elderly Patients’ Satisfaction towards Out-patient Pharmacy Services in General Hospitals

STEP 2:
CONSTRUCTING THE SCOPE LAYER

Scope Layers consist of two categories:

1. System layer 2. Sub-system or Element layers

Each category is defined by its various types

SYSTEM Layer

1. Begin the 1^{st} layer with the **SYSTEM**
2. For the next layers, identify the **types** of the SYSTEM
3. Continue Step 2 until you exhaust the types of the SYSTEM

SUB-SYSTEM OR ELEMENT Layers

4. At the next layer after the last layer of **SYSTEM**, identify the **SUB-SYSTEMS/ELEMENTS** of the **SYSTEM**
5. For the next layers, identify the **types** of the SUB-SYSTEM
6. Continue Step 5 until you exhaust the types/applications of the SUB-SYSTEM
7. Repeat Steps 4 through 6 for the next elements depending on the depth you want

THE SCOPE LAYERS: AN EXAMPLE

General Topic

Factors Influencing the Performance of Motor Vehicles

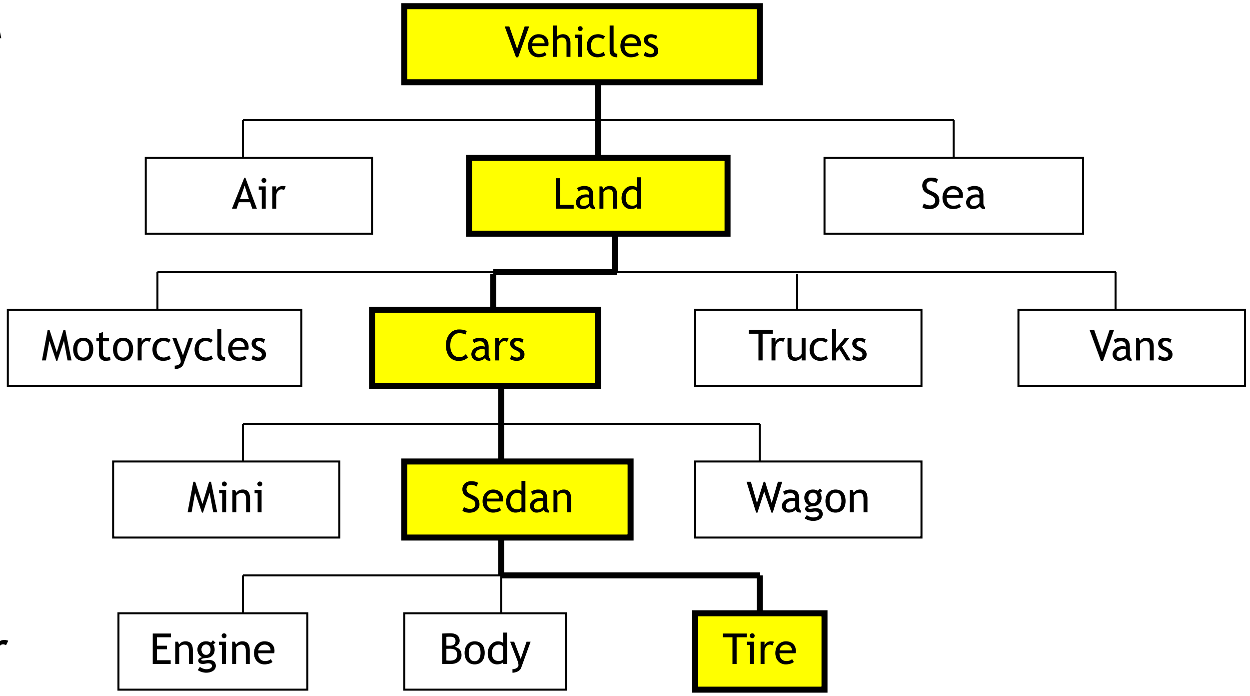
The SYSTEM

Type 1: Applications

Type 2: Vehicle Types

Type 3: Car Types

SUB-SYSTEMS or ELEMENTS of Sedan Cars



HOW TO CHOOSE AN OPTION: PROCEEDING TO THE NEXT LEVEL

Availability of Expertise	Availability of Facility	Required Time
Complexity of research	Novelty	Cost
Performance	Completeness of knowledge/ solution	Importance: Applicability, Impact, Urgency

THE IMPORTANCE OF SYSTEM LAYER

- The more detail the layers, the less the assumption, the better
- Every time there is a jump between the layers, an assumption is made (thus, a justification is required)
- One can choose any way he/she likes to arrange the sequence of layers
- However, Issues of the same theme should be in the same layer
- The Issues Layers indicate
  - The scope of critical reviews (what to get from the papers)
  - The focused issues (designation of issues)
  - The rationale in choosing issues to study (to move from one level to the next)
  - The Problem Statement

SAMPLE SYSTEMS AND ELEMENTS

BIL	SYSTEM	ELEMENTS	SUB-ELEMENTS
1.	Car	Body; Engine; Tire	Body: Hood, Bonnet, Chassis Engine: Piston, Shafts,
2.	Computer	Software; Casing; Electronics;	Software: Operating Systems, Applications
3.	Human	Physical body, Mind; Soul	Physical: Head, Limbs, Torso etc
4.	School	Buildings; Teachers; Students; Administrators; Rules; Curriculum;	Buildings: Admin., Classrooms, Halls
5.	Tourism Industry	Tourists; Hotels; Transportations; Rules	Tourists: Local, Foreign

IDENTIFYING THE TYPES

- Types can be defined based on any categories:
  1. **Size**
  2. **Application/Usage**
  3. **Models**
  4. **Standards**
  5. **Technology**
  6. **Methods/Techniques**

NOTICE

System must be at the top, followed by its types

Sub-system must be at the lower layer, and followed by its types

One can choose any way he/she likes to arrange the sequence of the types within a System or a Sub-system. However, Issues of the same type should be in the same layer

The more detail the layers, the less the assumption, the better

Every time there is a jump between the layers, an assumption is made (thus, a justification is required as to the validity of the assumption)

One can choose any way he/she likes to arrange the sequence of layers

However, Issues of the same theme should be in the same layer (apply the layering checking mechanism)

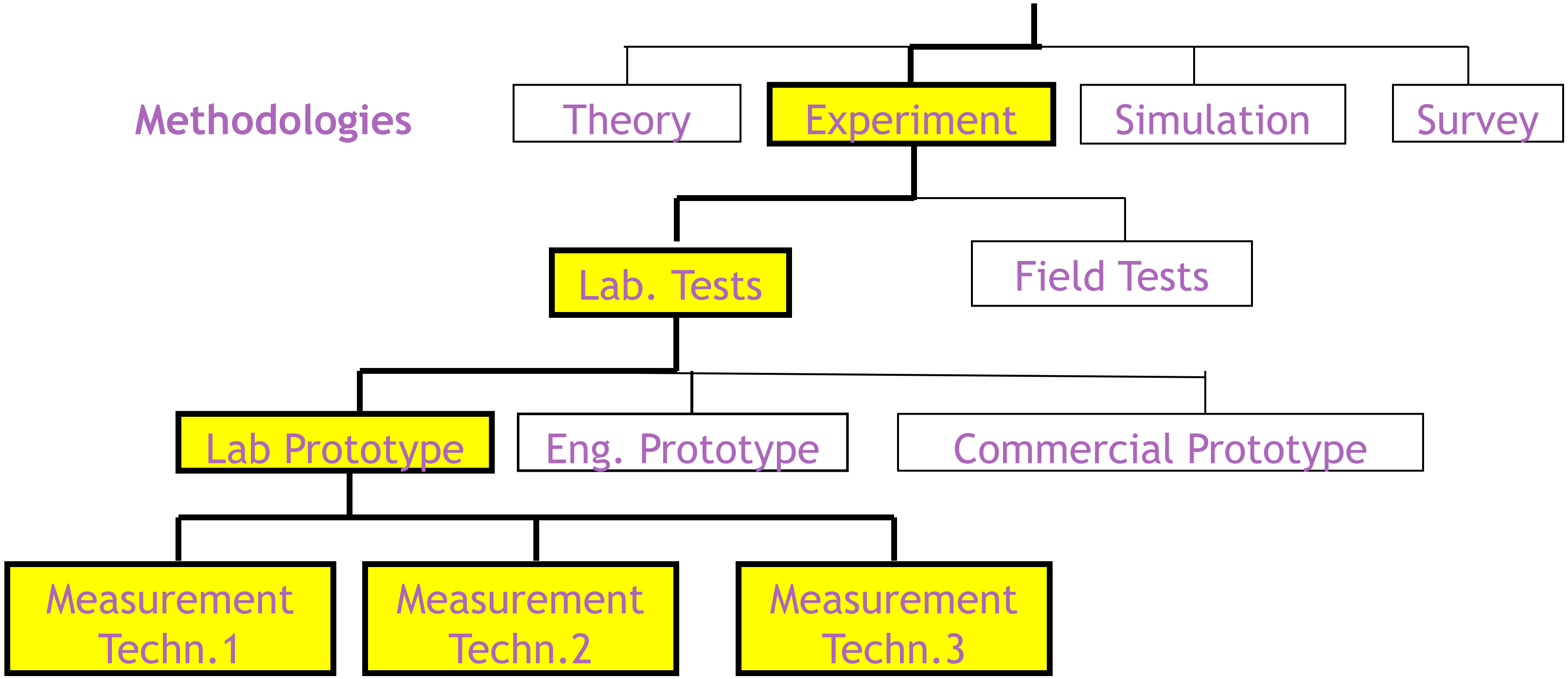
Perform extensive literature reviews

(RECAP):HOW TO CHOOSE AN OPTION: PROCEEDING TO THE NEXT LEVEL

- Availability of Expertise
- Availability of Facility
- Required Time
- Complexity
- Novelty
- Cost
- Performance
- Importance: Impact, Urgency

STEP 3:
THE METHODOLOGY LAYERS

8. For the 1st Methodology Layer, identify and choose the General Methods to use i.e Theory, Simulation, Experiment, Survey
9. For the subsequent layers, identify and choose the test environments, the level of prototypes, and the techniques to use



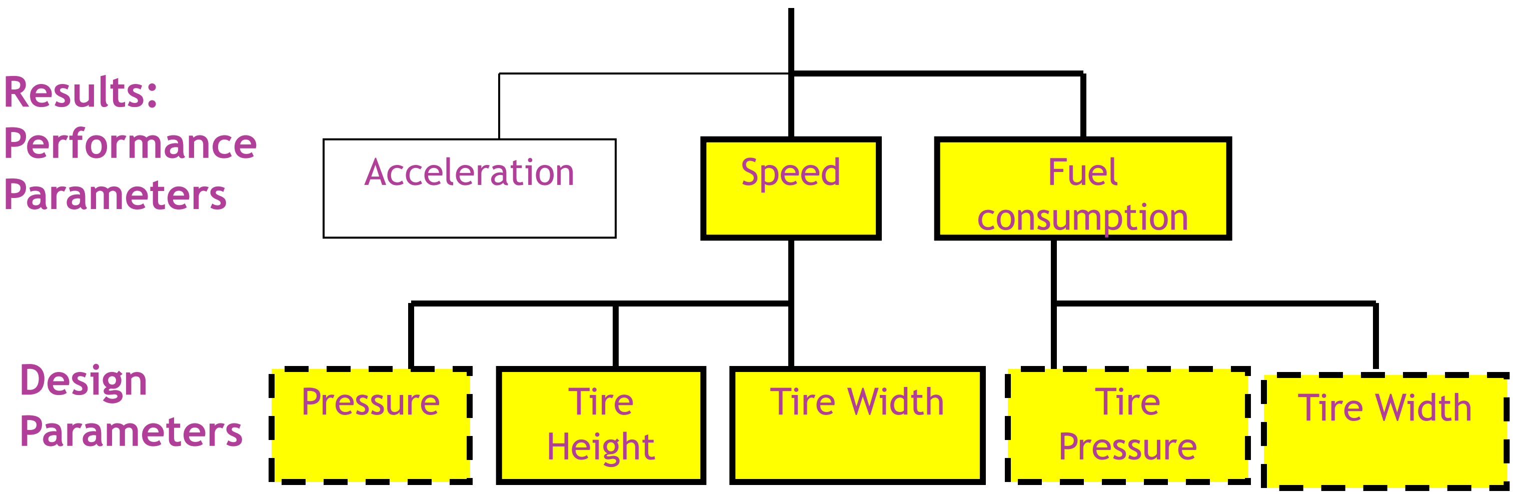
THE IMPORTANCE OF METHODOLOGY LAYER

- To define the specific methods adopted
- To designate which methods are of higher priorities
- To avoid ambiguities in the approach taken
- To reduce assumptions

STEP 4:
THE RESULT LAYERS

The **RESULTS** Layer consist of Performance Parameters(PP) followed by Design Parameters (DP).

10. Identify PPs. PPs (associated with the Outputs) – refer model
11. Identify DPs under the corresponding PPs. DPs are associated with the Internal and External factors
12. Identify the ones with higher priorities (dotted lines)



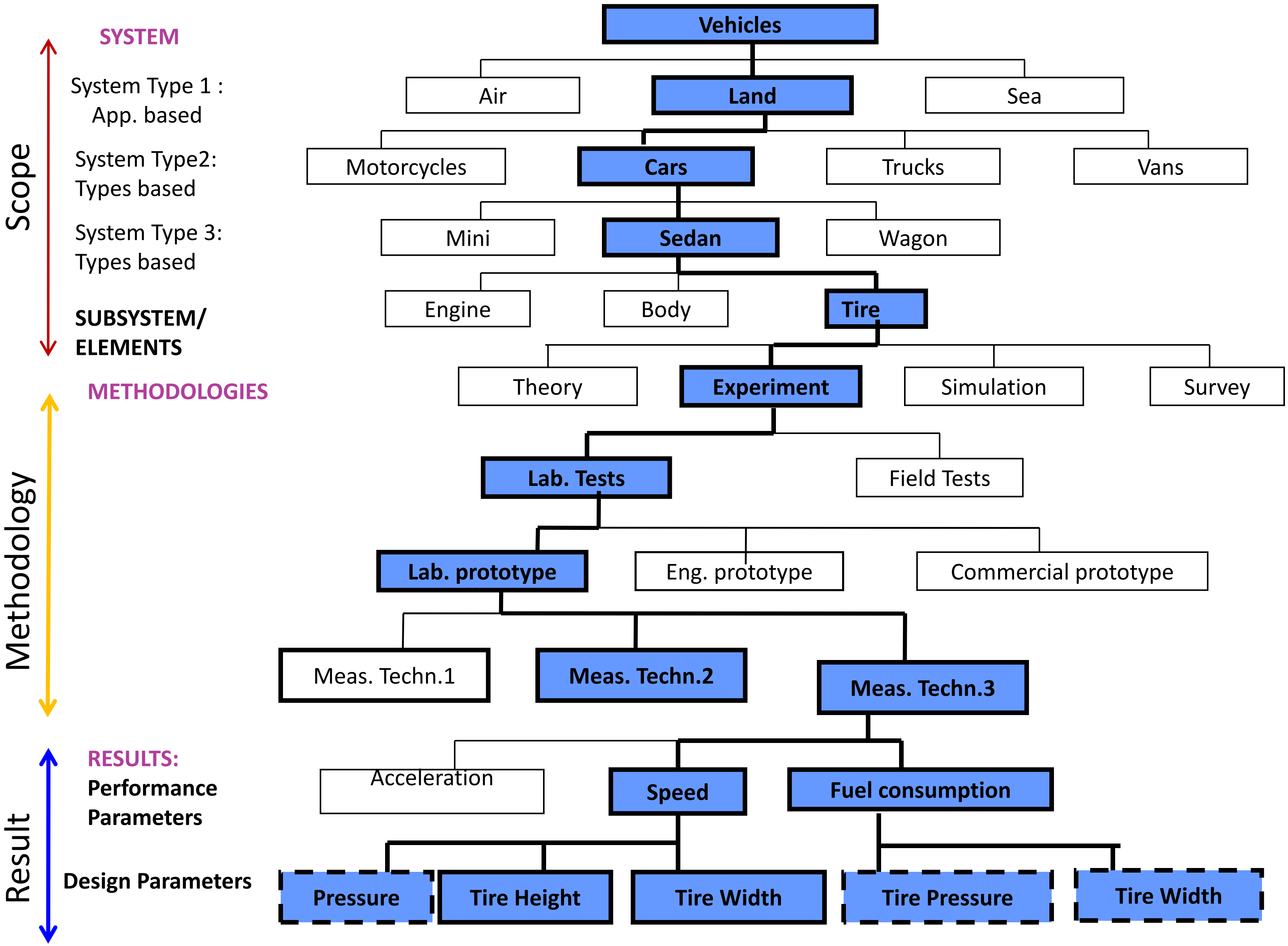
THE IMPORTANCE OF RESULT LAYER

The Results Layers help in:

1. Identifying the expected results, and how many of them
2. Setting the priorities and designation of results
3. Identifying the possible analyses e.g. comparative studies
4. Organizing reports/thesis/papers
5. Designating sub-projects

GENERAL TITLE

Factors Influencing the Performance of Motor Vehicles



EVALUATING A K-CHART™

- Is the any jump?
- What assumptions have been made?
- Are the assumptions justifiable?
- Are the options at each layer exhaustive?
- How the options are made to move to the next layers?
- What could be potential titles?
- How many results are expected?
- How many good papers can be expected?

CONCLUSION

- Most existing tools are suitable for macro level research planning
- K-Chart™ is a micro-level research management & planning tool
- It is useful for planning, execution and completion of a research project
- K-Chart™ is constructed based on Tree Diagram concept; focus and expand processes

THANK YOU